

Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/Alan/Static · 35 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/Alex/Static · 35 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/Georgia/Static · 35 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/Henry/Static · 35 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/Jestin/Static · 36 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/Joselyn/Static · 30 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/Josh/Static · 35 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/Nat/Static · 37 trials
Total training paths	8 · 278 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/Marcus/Static
Report output dir	/home/jestin/ThesisRepo/ML/saved_reports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	176 channels
Classes	7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 5.0 Hz · order 4 · fs 30.0 Hz
Normalisation	standard
Feature mode	stats → feature dim 1408
Test size · seed · CV folds	0.2 · random_state=42 · 10-fold stratified
Sample counts	train (post-aug)=444 · test=56

Augmentation (training only)

Enabled	yes
Copies per train trial	1
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.8, 1.2)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (10-fold)	CV std	Hold-out test acc
Random Forest	0.9932	0.0145	0.9643
SVM (RBF)	0.9955	0.0136	0.9464
SVM (Linear)	0.9977	0.0068	0.9464
Logistic Regression	0.9955	0.0136	0.9464
Voting Soft (SVM+RF+KNN+LogReg)	0.9955	0.0136	0.9286
Stacking (SVM+RF+KNN+LogReg)	0.9955	0.0136	0.9286
KNN	0.9075	0.0435	0.8571
LDA	0.9212	0.0232	0.3036

Per-class hold-out F1

Class	Random Forest	SVM (RBF)	SVM (Linear)	Logistic Regression	Voting (soft)	Stacking	KNN	LDA	Support
Double_Closed_Fist	0.947	1.000	0.947	0.900	0.900	0.900	0.857	0.414	9
Double_Nothing	0.833	0.875	0.923	0.923	0.833	0.833	0.727	0.000	7
Double_Okay	1.000	0.857	0.857	0.857	0.857	0.857	0.667	0.000	8
Double_Open_Palm	1.000	0.941	0.889	0.941	0.941	0.941	0.941	0.588	8
Double_Phone	1.000	1.000	1.000	1.000	1.000	0.941	0.857	0.200	8
Double_Pistol	0.941	1.000	1.000	1.000	0.941	1.000	0.889	0.296	8
Double_Spiderman	1.000	0.933	1.000	1.000	1.000	1.000	1.000	0.182	8
macro avg	0.960	0.944	0.945	0.946	0.925	0.925	0.848	0.240	56

External test (NewTestData)

Algorithm	External test acc
Random Forest	0.9714
Voting Soft (SVM+RF+KNN+LogReg)	0.9429

